Supplementary File S2: PRISMA-ScR Checklist

Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist

Manuscript Title: Large Language Models in Clinical Nutrition: A Scoping Review of Performance, Safety, and Implementation

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Target Journal: European Journal of Clinical Nutrition (EJCN)

PRISMA-ScR Checklist Items (1-22)

Item #	Section	PRISMA-ScR checklist item	Reported in
1	Title	Identify the report as a scoping review.	Title
2	Abstract	Provide a structured summary (background; objectives; eligibility criteria; sources; charting methods; results; conclusions).	Abstract (unstructured per EJCN Review Article format).

3	Introduction	Describe the rationale for the review and why a scoping approach is appropriate.	Introduction
4	Introduction	Provide an explicit statement of the questions and objectives being addressed.	Research Objectives
5	Methods	Indicate whether a protocol exists and any registration information.	Methods: Protocol/registration statement
6	Methods	Specify eligibility criteria (years, language, publication status) with rationale.	Methods: Inclusion and Exclusion Criteria
7	Methods	Describe information sources and the date of the most recent search.	Methods: Search Strategy
8	Methods	Present the full electronic search strategy for at least one database so it can be repeated.	Supplementary_File_S3_Search_Strings.xlsx + Supplementary_File_S3_Complete_Search_Strategy.docx
9	Methods	State the process for selecting sources of evidence (screening and eligibility).	Methods: Screening and Selection
10	Methods	Describe methods of charting data from included sources (forms/fields, independent charting, etc.).	Methods: Data extraction/charting
11	Methods	List and define all variables for which data were sought and	Methods + Supplementary_Table_S1_Included_Studies.xls

		any assumptions/simplifications made.	x headers
12	Methods	If done, provide a rationale and methods for critical appraisal of individual sources.	Not performed (scoping review); stated in Methods
13	Results	Describe methods of handling and summarizing the charted data.	Results: Thematic synthesis approach
14	Results	Give numbers screened/assessed/included and reasons for exclusions (ideally via flow diagram).	Results: Literature Search Outcomes + Figure 1
15	Results	Present characteristics of the included sources and provide citations.	Supplementary_Table_S1_Included_Studies.xlsx
16	Results	If done, present data on critical appraisal within sources (see item 12).	Not performed (scoping review); stated in Methods
17	Results	For each included source, present relevant charted data that relate to the objectives.	Results + Supplementary_Table_S1_Included_Studies.xlsx
18	Results	Summarize/present charting results as they relate to objectives.	Results (thematic synthesis)
19	Discussion	Summarize main results, link to objectives, and discuss relevance to key groups.	Discussion
20	Discussio	Discuss limitations of the scoping review	Limitations

	n	process.	
21	Discussion	Provide overall interpretation with implications/next steps.	Conclusions
22	Funding	Describe sources of funding for the scoping review and the role of funders.	Funding/COI statements + Supplementary_Table_S1_Included_Studies.xlsx

PRISMA Flow Summary (Figure 1)

- Records identified: 8,247
- Duplicates removed: 2,415
- Records screened: 5,832
- Records excluded (title/abstract): 5,520
- Full-text reports assessed: 312
- Full-text reports excluded: 259
- Sources included: 53

Supplementary File S3: Complete Search Strategy

Large Language Models in Clinical Nutrition: A Scoping Review of Performance, Safety, and Implementation

This file documents the complete electronic search strategy used for this scoping review, including databases searched, date limits, and where to find the full search strings.

Databases and Coverage

We searched the following sources for records published between January 1, 2020 and September 30, 2025; a bridge search updated the evidence base through December 22, 2025:

- PubMed/MEDLINE
- Web of Science Core Collection
- Scopus
- IEEE Xplore
- ACM Digital Library
- Google Scholar

Language restriction: English.

Full Search Strings (All Queries)

All 65 database-specific Boolean queries (exact strings, with limits and notes) are provided in:

- Supplementary_File_S3_Search_Strings.xlsx

Bridge search (Oct-Dec 2025): database updates and supplementary screening were performed to capture late-2025 publications, yielding four additional included sources.

Search Output Summary (PRISMA)

Across databases, the combined export contained 8,247 records prior to deduplication. After deduplication (2,415 duplicates removed), 5,832 unique records were screened by title/abstract, and 5,520 were excluded. Full-text assessment was conducted for 312 reports; 259 were excluded at full text, and 53 sources met eligibility criteria and were included in the scoping review.

Supplementary Table S3. Ethical Frameworks for AI in Clinical Nutrition

Ethical Domain	Key Concern	Implementation Strategy	Regulatory Reference
Bias and Fairness	Algorithmic perpetuation of racial, gender, and socioeconomic health disparities (e.g., Obermeyer et al., 2019)	Diverse Training Data: Enforce inclusion of underrepresented datasets. Bias Auditing: Regular audits for disparate impact across demographic subgroups. Cultural Adaptation: Fine-tuning models on regional dietary patterns.	EU AI Act: Requirements for high-risk AI systems regarding data governance (Article 10). WHO Guidance: "Protecting human autonomy" and "Ensuring equity."
Transparency and Explainability	"Black box" nature of deep learning models obscuring the rationale behind dietary recommendations.	Explainable AI (XAI): Implement local interpretable model-agnostic explanations (LIME) or SHAP values. Source Attribution: RAG systems citing specific guidelines for every claim. Confidence Scoring: Displaying uncertainty metrics for generated advice.	FDA SaMD: Transparency requirements for clinical decision support software. GDPR: Right to explanation (Articles 13-15).
Privacy and Data Security	Handling sensitive patient health data (PHI) in cloud-based model inference.	Local Inference: Deployment of quantized models on edge devices (privacy-first). De-identification: Robust	HIPAA: Protected Health Information standards. GDPR: General Data Protection Regulation

		anonymization protocols before cloud processing. Federated Learning: Training across decentralized institutions without sharing raw data.	requirements.
Accountability and Liability	Unclear legal liability for adverse outcomes resulting from AI errors (hallucinations).	Human-in-the-Loop (HITL): Mandatory dietitian review for high-stakes recommendations. Liability Disclaimers: Explicit statements that AI provides information, not medical advice. Insurance Models: Developing malpractice frameworks covering AI usage.	EU AI Act: Accountability framework for providers and users. Tort Law: Emerging precedents on software liability.
Clinical Safety	Risk of potentially fatal advice (e.g., allergen errors) or encouraging eating disorders.	Red Teaming: Adversarial testing for dangerous outputs before deployment. Guardrails: Hard-coded rules blocking generation of advice on severe medical conditions (e.g., anorexia). Monitoring: Post-market surveillance of adverse events.	FDA: Post-market surveillance requirements. ISO/IEC 42001: AI Management System standards.
Sustainability	Environmental impact of training large models versus	Efficient Architectures: Prioritizing "Small	EU AI Act: Environmental reporting

	health benefits.	Language Models" (SLMs) for specific tasks. Green AI: Reporting carbon footprint of model training and inference.	obligations for general-purpose AI models.